

Problem 1.5

When interest is calculated on the original principal only it is called *simple interest*. Accumulated interest from prior periods is not used in calculations for the following periods. In this case, the amount value A , the principal P , the period of payment t , and the annual interest rate i are related by the formula

$$A = P(1 + it).$$

At what rate will \$500 accumulate to \$615 in 2.5 years?

Solution

Using the simple interest formula,

$$615 = 500(1 + i \cdot 2.5).$$

Divide both sides by 500:

$$\frac{615}{500} = 1 + 2.5i.$$

Then,

$$\frac{615}{500} - 1 = 2.5i.$$

Since

$$\frac{615}{500} = 1.23,$$

we get

$$1.23 - 1 = 2.5i$$

so

$$0.23 = 2.5i.$$

Therefore,

$$i = \frac{0.23}{2.5} = 0.092.$$

Hence, the required annual interest rate is

$$\boxed{9.2\%}.$$